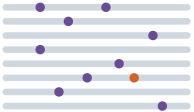
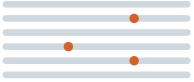
LIBRARIES
nanopore, Illumina or Sanger

AID-competent animals



AID-null controls (two or more)





nanopore Illumina Sanger

Sample sheet

Each file, or each Sanger clone, is assigned to an animal, a group, and control or not. Order and well stay with every flag.

READ PREPARATION
every read is gated

Gates

Reads over a length set for each locus (Methods), or averaging more than 0.01 errors per base, are dropped. Sanger: the longest stretch at Q ≥ 20.

Competitive alignment

Each read is aligned to the target, its paralogue and any contaminant at once, and goes to its best match. A paralogue can be analysed as a target of its own.



target paralogue other

Read accounting

The fate of every delivered read, with its reason, for every library (Figure S1).

AUTOMATED ANALYSIS
calibrated by the AID-null controls

Background

Measured in the controls, per position, change, strand and base quality, and inflated by one factor fitted leaving each control out in turn.

Candidate test

A Poisson tail over every depth of evidence, against that background. A change in half the reads or more is set aside as likely germline (Sanger: per sequencing batch).

Threshold set by the controls (target–decoy)

Each control is called against the others, giving decoy p-values. The threshold is the largest at which the false calls the decoys predict are at most 5% of the calls kept.

Sanger: every substitution at Q ≥ 20 counts; the AID-null clones measure the error, and each animal is told how many of its calls that background explains.

Bookkeeping checks (reported, never applied)

Copied traces and duplicate clones, as groups; clones shared between animals beyond chance; mutated control clones; label mismatches; load outliers. A trace filed under two animals on purpose is kept under both; an accidental copy is counted once.

RESULTS
per animal, then per group

Mutation profiles

unique-variant density
mutation frequency
12-type spectrum, corrected for base composition
AID (WRC/GYW) and Polη (WA/TW) hotspots
position along the amplicon
indels, clones and mutation trees
calls the background explains (Sanger)

Group tests

Welch t on $\log_{10}$ density, one value per animal, so clones of one lineage cannot stand in for independent mice.



case control

● mutation ● sequencing error